

Phase-2 Demo Writeup — Exam Ready Answers

Project: AI-Powered Student Performance Tracker
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1. Problem Statement

Project Title: AI-Powered Student Performance Tracker

Explanation:

Our project is a full-stack web application that uses Artificial Intelligence to **proactively predict** a student's exam score and burnout risk level **before** the exam happens.

Currently, colleges follow a **reactive approach** — teachers and parents find out a student is struggling only **after** the exam results are declared. By that time, it is too late to help the student.

Our system solves this by collecting **14 daily lifestyle and academic habits** from students (like study hours, sleep hours, attendance, screen time, mental health score, caffeine intake, gaming hours, etc.) and feeding them into a **Random Forest Machine Learning model**. The model instantly predicts:

- Predicted Exam Score** (0 to 100%)
- Risk Level** — Low, Medium, or High

But prediction alone is not enough. Students and teachers need to know **WHY** the AI gave that result. So we integrated **SHAP (SHapley Additive exPlanations)**, an Explainable AI technique. It shows exactly which habit is helping or hurting the score (e.g., "-5% due to low sleep", "+12% due to high study hours").

The system has two portals:

- Student Portal** → Students log habits, view their AI prediction, and see SHAP explanations.
- Admin Portal** → Teachers view all students, filter by risk level, upload CSV files for batch predictions, and prepare for counseling.

In short: Our project shifts education from *"finding out a student failed after the exam"* to *"detecting the student is at risk weeks before the exam and helping them fix their habits."*

2. Requirements

Hardware Requirements

Component	Specification
Processor	Intel Core i3 or higher
RAM	8 GB minimum (16 GB recommended for ML training)
Storage	500 MB for code + 2 GB for dependencies
Network	Stable Internet (for cloud database & deployment)
Display	1366×768 minimum resolution

Software Requirements

Software	Version / Details
Operating System	Windows 10/11, macOS, or Linux
Programming Language (Backend)	Python 3.11+
Programming Language (Frontend)	JavaScript (ES6+)
Backend Framework	Django 5.0, Django REST Framework
Frontend Framework	React.js 18 (via Vite 5)
ML Libraries	Scikit-Learn, Pandas, SHAP, Joblib
Database	PostgreSQL (hosted on Supabase)
Styling	Tailwind CSS v4, Framer Motion
Email Service	Brevo SMTP (for OTP verification)
Version Control	Git + GitHub
IDE	VS Code
Deployment	Vercel (Frontend), Render (Backend)

3. Project Description — Module-wise Working

Our project has **5 main modules**:

Module 1: Authentication & Security Module

- Users register by entering their email. The system sends a **6-digit OTP** to their email via Brevo SMTP.
- User must enter correct OTP to verify their account.
- After verification, user can login. The system generates a **JWT (JSON Web Token)** for secure session management.
- Two roles exist: **Student** and **Admin**. Each role sees a different dashboard.
- Password reset is also done via Email OTP.

Module 2: Student Data Entry Module

- After login, the student fills out a form with **14 habit metrics**:
 - Study Hours, Self-Study Hours, Online Class Hours
 - Attendance Percentage, Class Participation

- Sleep Hours, Exercise Minutes
 - Social Media Hours, Gaming Hours, Total Screen Time
 - Caffeine Intake, Mental Health Score (1-10)
 - Part-Time Job (Yes/No), Upcoming Deadlines count
- The form has **validation** — for example, sleep hours cannot be negative or more than 24.

Module 3: Machine Learning Prediction Module (Core AI Engine)

- When the student submits the form, the backend loads two pre-trained **Random Forest** models:
 1. **RandomForestRegressor** → Predicts the exact exam score (0-100%).
 2. **RandomForestClassifier** → Predicts Risk Level (Low / Medium / High).
- The models were trained on a **custom 10,000-record dataset** created by merging two Kaggle datasets.
- The models are stored as `.pkl` files using **Joblib** and loaded instantly — no external API calls needed.

Module 4: Explainable AI (SHAP) Module

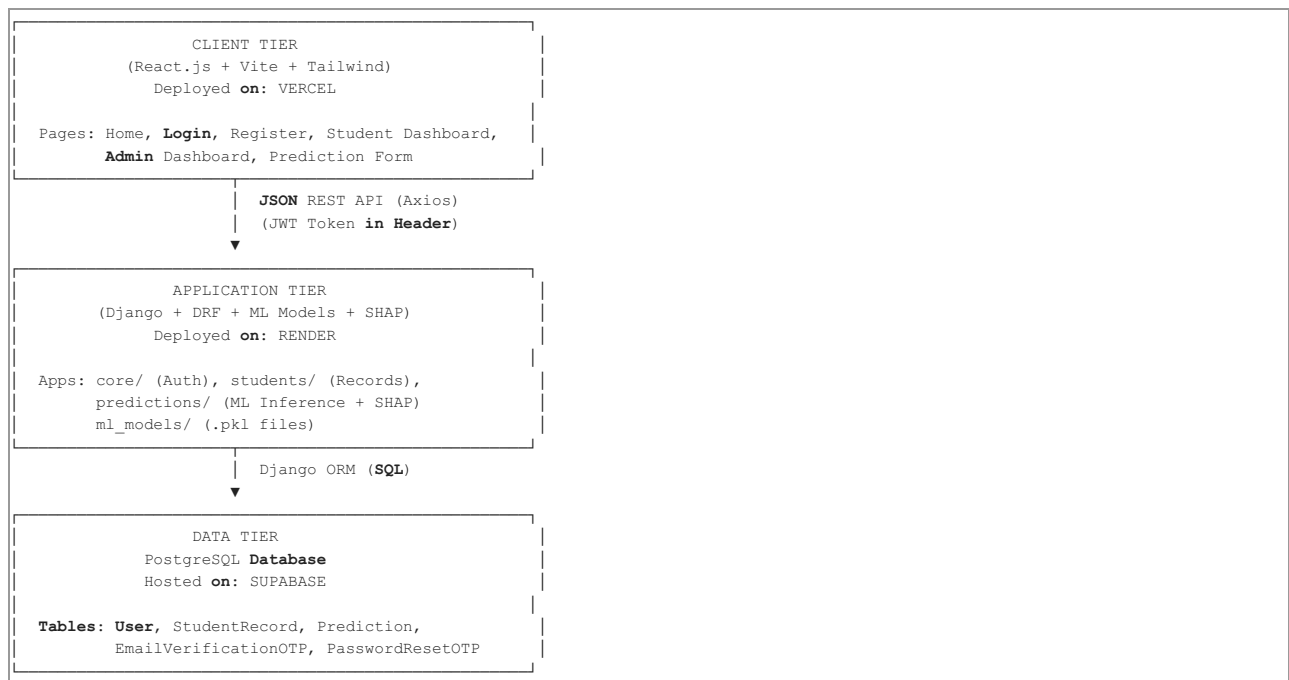
- After prediction, the system runs **SHAP TreeExplainer** on the same input.
- SHAP calculates the **marginal contribution** of every single feature to the final score.
- The result is a list showing: which habit helped the score (+ve impact) and which hurt it (-ve impact).
- Example output: "sleep_hours: +8.5" means good sleep added 8.5% to the score. "gaming_hours: -6.2" means excessive gaming reduced the score by 6.2%.
- This is displayed on the dashboard as **color-coded bar charts** (green = positive, red = negative).

Module 5: Admin Command Center Module

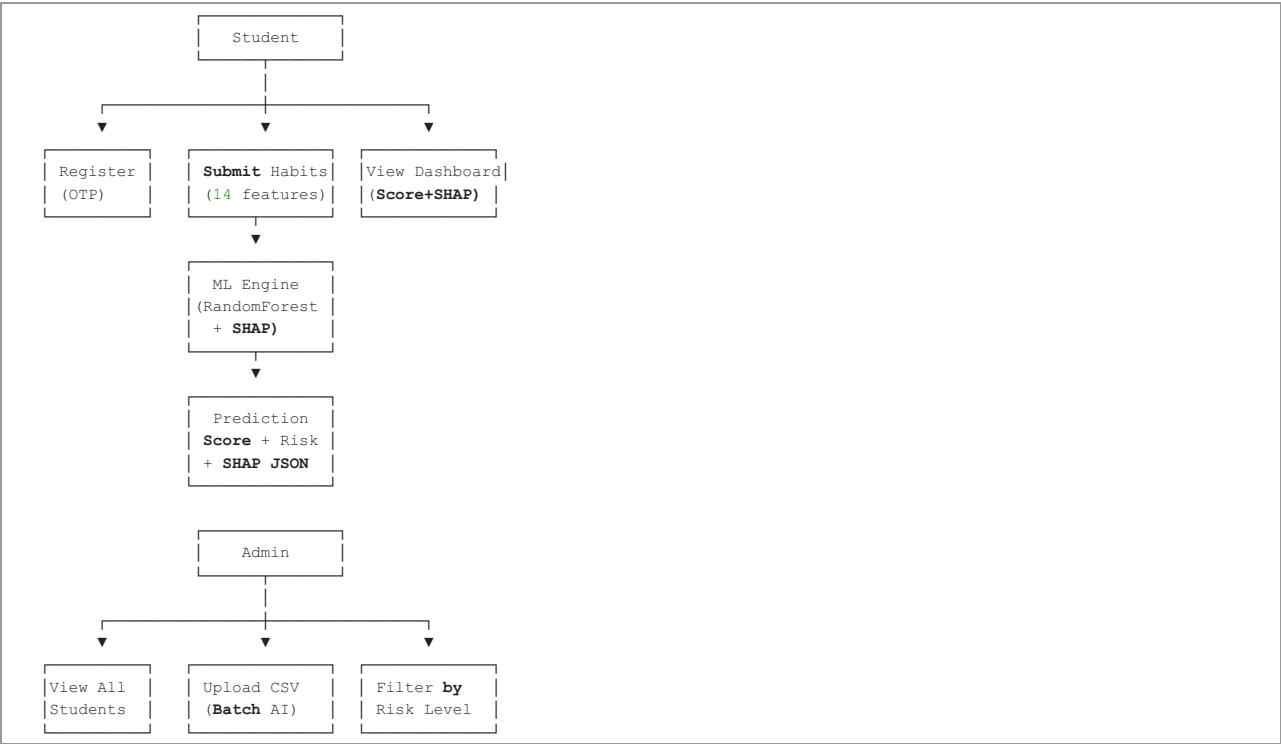
- Admins can see a **table of all students** with their predicted scores and risk levels.
- They can **filter by risk level** (e.g., show only "High Risk" students for counseling).
- **Batch CSV Upload:** Admin uploads a `.csv` file with data of 100+ students → the system runs predictions on all of them **instantly** and saves results to the database.
- Admin can also enter the **actual exam score** later to compare with the AI prediction (accuracy tracking).

4. System Design

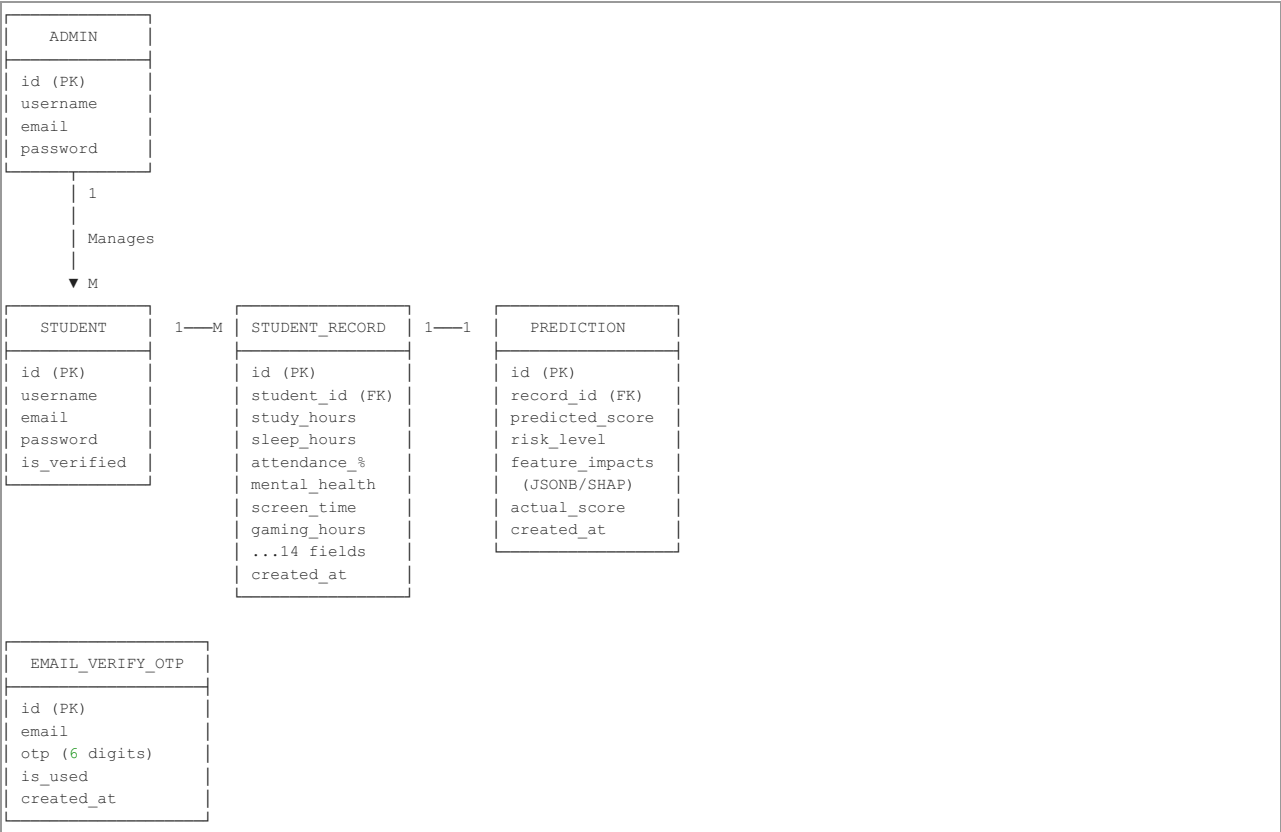
4.1 Architecture Diagram (3-Tier Decoupled Architecture)



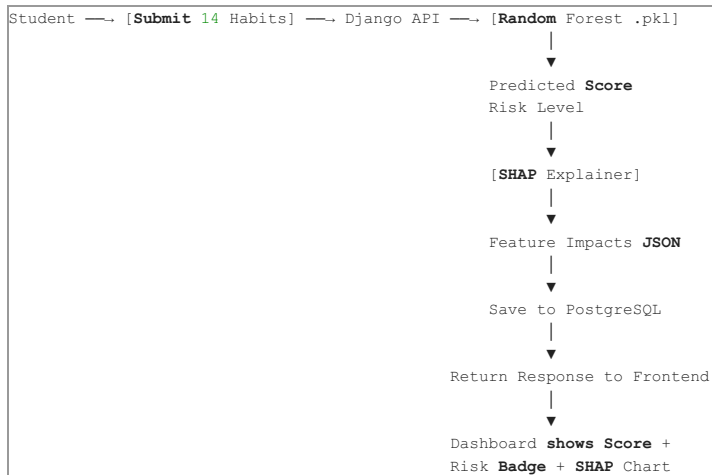
4.2 Use Case Diagram (Actors & Actions)



4.3 ER Diagram (Entity Relationship)



4.4 Data Flow Diagram (Level 1)



5. Conclusion

The **AI-Powered Student Performance Tracker** successfully demonstrates that Machine Learning can be integrated into education to create a **proactive early-warning system** for struggling students.

Key achievements of this project:

1. We trained **Random Forest ML models** on a custom 10,000-record dataset to predict exam scores and risk levels with high accuracy.
2. We integrated **SHAP Explainable AI** so the system doesn't just predict — it **explains** exactly which habits are helping or hurting the student.
3. We built a **full-stack web application** using React.js (frontend) and Django REST Framework (backend), deployed live on Vercel and Render.
4. The **Admin batch CSV feature** allows teachers to process an entire classroom of students in seconds.
5. The system is **secure** with JWT authentication and Email OTP verification.

In simple terms: Instead of finding out a student failed *after* the exam, our system detects the student is at risk *weeks before* the exam and tells them exactly what habits to fix.

Future Scope:

- Real-time email/SMS alerts to parents when a student enters "High Risk"
- Mobile app version using React Native
- Integration with college LMS (like Moodle) for automatic data collection

Live Demo URL: <https://ai-powered-student-performance-tracker.vercel.app>
GitHub: <https://github.com/Madhav-P-2005/AI-Powered-Student-Performance-Tracker>